

IT23431 – MongoDB Essentials Complete Notes

UNIT I – Introduction to MongoDB

Introduction to NoSQL Databases

Exam-oriented notes covering Introduction to NoSQL Databases. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

Types of NoSQL Databases: Key-Value, Document, Column-Family, Graph

Exam-oriented notes covering Types of NoSQL Databases: Key-Value, Document, Column-Family, Graph. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

SQL vs NoSQL

Exam-oriented notes covering SQL vs NoSQL. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

MongoDB Features, Architecture and Ecosystem

Exam-oriented notes covering MongoDB Features, Architecture and Ecosystem. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

MongoDB Atlas, Compass and Shell

Exam-oriented notes covering MongoDB Atlas, Compass and Shell. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

Installation and Setup

Exam-oriented notes covering Installation and Setup. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

UNIT II – CRUD Operations

Databases and Collections

Exam-oriented notes covering Databases and Collections. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

insertOne(), insertMany()

Exam-oriented notes covering insertOne(), insertMany(). Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

find(), Filtering and Projection

Exam-oriented notes covering find(), Filtering and Projection. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

updateOne(), updateMany(), replaceOne()

Exam-oriented notes covering updateOne(), updateMany(), replaceOne(). Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

deleteOne(), deleteMany()

Exam-oriented notes covering deleteOne(), deleteMany(). Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

UNIT III – Data Modeling in MongoDB

BSON Format

Exam-oriented notes covering BSON Format. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

Document Structure

Exam-oriented notes covering Document Structure. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

Embedded and Referenced Documents

Exam-oriented notes covering Embedded and Referenced Documents. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

One-to-One, One-to-Many, Many-to-Many Relationships

Exam-oriented notes covering One-to-One, One-to-Many, Many-to-Many Relationships. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

Indexing: Single, Compound, Text and Geospatial

Exam-oriented notes covering Indexing: Single, Compound, Text and Geospatial. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

UNIT IV – Aggregation Framework

Aggregation Pipeline

Exam-oriented notes covering Aggregation Pipeline. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

\$match, \$group, \$project

Exam-oriented notes covering \$match, \$group, \$project. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

\$sort, \$limit, \$skip

Exam-oriented notes covering \$sort, \$limit, \$skip. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

\$lookup

Exam-oriented notes covering \$lookup. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

\$bucket and \$facet

Exam-oriented notes covering \$bucket and \$facet. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

UNIT V – MongoDB Integration

MongoDB Driver

Exam-oriented notes covering MongoDB Driver. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

MongoClient, MongoDBDatabase, MongoCollection

Exam-oriented notes covering MongoClient, MongoDBDatabase, MongoCollection. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

Java Integration

Exam-oriented notes covering Java Integration. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

CRUD using Java

Exam-oriented notes covering CRUD using Java. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.

Best Practices

Exam-oriented notes covering Best Practices. Include definitions, features, examples, advantages, applications, and important points for Anna University examinations.